

PROFILE

Final-year Mechanical Engineering student at RMIT University (graduating December 2026) with hands-on experience across computational fluid dynamics, experimental flow measurement, control systems and finite element analysis. Capstone work validates ANSYS Fluent simulations against PIV measurements on a purpose-built flow rig. Two years of part-time work at Coles alongside full-time study, bringing reliability, safety awareness and strong teamwork. Seeking a graduate mechanical engineering role in Melbourne.

EDUCATION

Bachelor of Engineering (Mechanical Engineering) (Honours)
RMIT University, Melbourne

[Start year] – Dec 2026

- [GPA: X.X / 4.0 — include if ≥ 3.0, otherwise remove]
- Relevant courses: Fluid Mechanics, Thermodynamics, Heat Transfer, Mechanics of Solids, Dynamics & Control, Engineering Design, Computational Engineering [edit to match your transcript]

ENGINEERING PROJECTS

Capstone — Closed-Loop PIV Flow Rig with Stenosis Inserts

2026

- Designed and built a closed-loop water flow rig with a transparent S-bend test section and interchangeable stenosis inserts to study curved, constricted flow.
- Measured velocity fields using particle image velocimetry (PIV) and developed matching ANSYS Fluent CFD models for validation.
- [Key result — e.g. CFD agreed with PIV within X% at Re = Y; quantified recirculation zone downstream of stenosis.]

ROS2 PID Line-Following Robot

[Year]

- Implemented a ROS2 sensing → PID control → velocity command pipeline in Python and tuned gains on physical hardware.
- [Key result — e.g. reduced lap time by X% / eliminated oscillation on tight curves.]

Autonomous Vehicle Simulation (MATLAB)

[Year]

- Developed line-tracking, obstacle-avoidance and smart-parking behaviours on a kinematic vehicle model in MATLAB.
- [Key result.]

7.2 kW Residential Solar PV System (NREL SAM)

[Year]

- Modelled system performance and financials using local weather data, component specifications and loss assumptions. [Annual yield ___ kWh, payback ___ years.]

FEA Pedal Design

[Year]

- Performed static structural FEA with mesh-convergence checks and iterated geometry for strength-to-weight. [FoS ___, mass reduced ___%.]

WORK EXPERIENCE

Team Member — Coles Supermarkets

[Month] 2024 – Present

[Store name], Melbourne · Part-time, 2 years

- Worked reliably in a fast-paced, high-volume environment while studying full-time engineering.
- Followed workplace health & safety procedures, including safe manual handling and equipment use.
- Met time-critical targets for stock replenishment and store presentation, prioritising tasks across the shift.
- Resolved customer enquiries and communicated effectively with team members and supervisors.
- [Any extra responsibility — e.g. trained new team members, trusted with opening/closing, department lead, recognition award.]

TECHNICAL SKILLS

Simulation	ANSYS Fluent (CFD), ANSYS Mechanical (FEA), MATLAB, Simulink, NREL SAM
CAD & Design	SolidWorks, engineering drawings, design for manufacture [+ other CAD]
Controls & Code	ROS2, Python, PID control, Linux [+ C++/Arduino?]
Experimental	Particle image velocimetry (PIV), rig design & build, data processing
Professional	Technical report writing, team project delivery, WHS awareness, Microsoft Office

Referees available on request.